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Organic amendment-induced reducing conditions facilitate mercury methylation in paddy soils: Insights from the iron/sulfur reduction

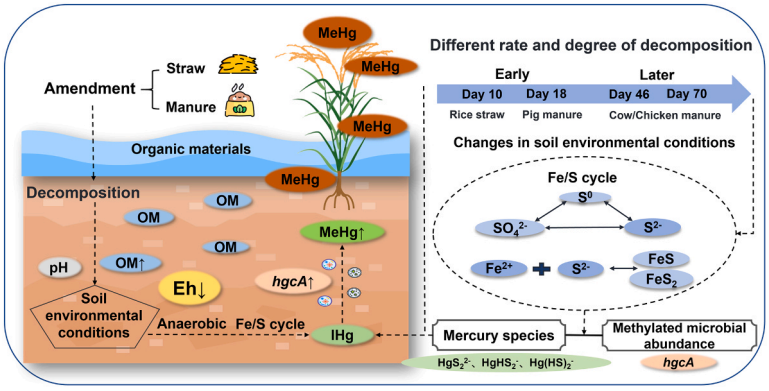
Juan Wang [a], Jingwen Yang [a], Yuping Xiang [a], Jinping Xue [b,1], Xiaosong Tian [c], Tao Jiang [a], Yongmin Wang [a,*], Dingyong Wang [a]

[a] College of Resources and Environment, Southwest University, Chongqing 400715, China
[b] Universite de Pau et des Pays de l'Adour, CNRS, Institut des Sciences Analytiques et des Physico-Chimie pour l'Environnement et les Matériaux (IPREM), Pau 64000, France
[c] College of Resources and Safety, Chongqing Vocational Institute of Engineering, Chongqing 402260, China

HIGHLIGHTS

- Methylmercury peaks early in rice straw and pig manure, late in cow and chicken manure.
- Organic amendments boost Hg methylation via Eh decline-induced Fe and S reduction.
- Late-phase methylmercury levels are linked with Fe/S precipitation and DOC decrease.

GRAPHICAL ABSTRACT



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ABSTRACT

Organic amendments in paddy fields, although agronomically beneficial, can potentially stimulate mercury (Hg) methylation due to distinct shifts in soil properties triggered by different types of amendments. Identifying the specific conditions that lead to heightened Hg methylation is thus essential for risk mitigation and sustainable management. Through incubation experiments with rice straw (RS), pig manure (PM), cow dung (CD) and chicken manure (CM), key drivers of Hg methylation were identified by linking methylation extent to amendment-induced dynamics in soil pH, Eh, Fe/S, and organic matter. Results revealed amendment-specific temporal dynamics governed by organic matter composition and reactivity. Rice straw and PM triggered a rapid Eh decline to –300 mV, concurrently accelerating Fe/S reduction and causing early methylmercury (MeHg) peaks on day 10 (7.51 ± 1.16 ng/g) and 18 (3.94 ± 0.32 ng/g), respectively. In contrast, CD and CM exhibited delayed Eh minimization and MeHg peaks (1.75 ± 0.58 and 12.94 ± 0.88 ng/g, respectively) until day 70. This differential MeHg production pattern was mainly driven by Eh-induced Fe/S reduction, which enhanced Hg

* Corresponding author.
E-mail address: wangym540@swu.edu.cn (Y. Wang).
[1] Present Address: Environmental Sciences Division, Oak Ridge National Laboratory, Oak Ridge, TN 37831, United States

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bioavailability through reductive iron-oxides dissolution and subsequent soluble Hg-S complexes formation, as well as increased abundance of Hg-methylating microbes. These findings highlight that organic materials decomposition kinetics regulate MeHg formation via reductive Fe-S cycling, supporting targeted contamination mitigation strategies in agriculture.

1. Introduction

Mercury (Hg) is a globally significant pollutant due to its high toxicity, environmental persistence, and capacity for bioaccumulation [1,2]. Methylmercury (MeHg), a potent neurotoxin formed predominantly through microbial methylation of inorganic Hg (IHg) under anaerobic conditions, is of particular concern [3,4]. Rice paddies, as widespread engineered wetlands, represent critical environments for Hg methylation, with rice grain serving as a major vector for human MeHg exposure, especially in regions where rice is a dietary staple [5–7]. Although the fundamental pathways of MeHg formation are established, substantial uncertainty remains regarding how common agricultural practices, particularly the application of organic materials, influence the biogeochemical drivers of Hg methylation in paddy systems [8–12].

Globally, agricultural organic materials such as crop straw and livestock manure, are increasingly applied to support sustainable agriculture by enhancing soil fertility, promoting nutrient recycling, and increasing carbon sequestration [13–16]. However, these amendments differ substantially in chemical composition (e.g., carbon/nitrogen ratios and organic matter content), nutrient availability, and decomposition pathways [15,17,18]. These differences are likely to induce contrasting effects on key soil properties, including redox potential (Eh), pH, organic matter dynamics, and iron (Fe) and sulfur (S) cycling, all of which potentially play crucial roles in Hg methylation [12,19–21].

Existing studies indicate that both straw and manure amendments can increase MeHg concentrations in paddy soils and rice grains [21–23]. However, the key soil properties through which different organic materials regulate the net production of MeHg remain poorly understood. Central to this knowledge gap is the interaction between microbially-driven organic matter decomposition and the biogeochemical cycling of iron and sulfur, a coupling that governs Hg mobility and bioavailability [24–26]. As microbial decomposition consumes oxygen, it can establish reducing conditions that facilitate the reductive dissolution of iron oxides, releasing absorbed Hg [18,24,25]. Concurrent sulfate reduction generates abundant sulfides, transforming Hg from the solid HgS into soluble and reactive polysulfide complexes such as HgS_2^{2-}, $HgHS_2^-$, and $Hg(SH)_2^0$, which enhance Hg bioavailability [27,28]. Conversely, these processes may also lead to Hg sequestration through the precipitation of metastable FeS and other insoluble phases [24,29]. This dual role, manifested as early-stage mobilization of Hg and late-stage sequestration, suggests a pronounced time-dependent effect on Hg methylation potential. The rate and extent of these processes are intimately linked to the decomposition kinetics and chemical characteristics of the organic amendments [21,22,30], yet a mechanistic understanding of these interactions remains inadequately quantified.

To address these challenges, this study employs an integrated approach combining incubation experiments with four representative organic amendments (rice straw, pig manure, cow dung, and chicken manure). We aim to (1) quantify the temporal dynamics of key soil properties, Hg speciation (based on PHREEQC model), and the abundances of Hg-methylating microorganisms across incubations of different organic amendments; (2) identify the predominant factors and their relative contributions to MeHg formation and further elucidate its potential mechanisms. Our results could provide essential data for the rational application of agricultural organic amendments and support the development of targeted management strategies to mitigate MeHg risk.

2. Materials and methods

2.1. Incubation experiment design

2.1.1. Materials preparation

The experimental paddy soil was obtained from the National Purple Soil Fertility and Fertilizer Effects Monitoring Station at Southwest University, Chongqing, China. The organic materials used in this study included rice straw (RS), a commercial organic fertilizer labeled as pig manure (PM, which was a typical 'manure-straw' mixture commonly used in agriculture), cow dung (CD) and chicken manure (CM). Following collection, both soil and organic materials were air-dried, ground, and sieved (2 mm) for incubation experiments. The soil was spiked with $HgCl_2$ solution, thoroughly homogenized, and subsequently aged for over 6 months, forming a final soil Hg concentration of 3.96 ± 0.22 μg/g. The basic property of the simulated paddy soil and organic materials are detailed in Table S1/S2.

2.1.2. Sample preparation and incubation

The simulated paddy soil cultivation was conducted with five treatments: a control (CK) and four organic materials treatments (RS, PM, CD, and CM). For each treatment, five replicate 50 mL anaerobic bottles were prepared by aliquoting 15 g of Hg-spiked soil. Organic materials were thoroughly incorporated into the soil at an application rate of 1 % (*w/w*). Subsequently, 40 mL of deionized water was added to each bottle, and the system was purged rapidly with high-purity N_2 for 10 min to remove excess O_2. The bottles were sealed, homogenized, and placed in a temperature-controlled incubator set to 25°C under dark conditions. Throughout the incubation period, anaerobic and dark conditions were maintained, and the bottles were shaken daily to ensure uniform mixing.

2.1.3. Sample collection and preservation

Destructive sampling was performed at seven time points corresponding to key stages of rice growth under flooded conditions: day 0 (4 h), day 10, day 18, day 33, day 46, day 70, and day 90 [31]. All sampling procedures were conducted in an anaerobic bag to maintain strict anoxia. Fresh soil samples were divided into three subsections: one portion was directly transferred to 50 mL centrifuge tubes under an N_2 atmosphere for soil elemental sulfur (S^0) analysis, another portion was immediately stored at –80°C for subsequent DNA extraction and *hgcA* gene (i.e., a key gene involved in Hg methylation) quantification via qPCR, and the remaining soil was centrifuged, freeze-dried, and homogenized to a fine powder (100 meshes) for further analysis. Additionally, the supernatant from the centrifuged fresh soil samples was filtered through 0.45 μm membranes (Millipore) under N_2 to obtain soil interstitial water. The interstitial water samples were then divided into two portions: one portion was preserved with 0.4 % *v/v* hydrochloric acid (GR) for the total Hg (THg) and MeHg determination, and another stored in amber glass vials at 4°C for DOM characterization and major ion (Fe^{2+}/Fe^{3+}, S^{2-}/SO_4^{2-}, Cl^-) analysis. All analyses were completed within one week.

2.2. Sample analyses

2.2.1. Soil properties determination

Soil pH and Eh were measured *in situ* using a calibrated portable multiparameter meter (HI-98130, HANNA®, Italy). Eh measurements were conducted inside an anaerobic bag to minimize oxygen

interference. Soil organic matter (SOM) was determined using the $K_2Cr_2O_7$ oxidation method [32]. Soil S^0 was extracted with methanol and quantified using high-performance liquid chromatography (LC-20A, Shimadzu Corporation, Kyoto, Japan) [33]. Fe^{2+} concentration in interstitial water was determined by colorimetric assay with ferrozine reagent [34], measured at 562 nm using a UV-Vis spectrophotometer (UV 1800PC, Shanghai Aoxi Scientific Instrument Co., Ltd., China). Total iron (TFe) was determined by the same procedure as Fe^{2+} after hydroxylamine hydrochloride reduction [34]. Fe^{3+} was calculated by subtracting Fe^{2+} from the TFe concentration. Sulfate (SO_4^{2-}) and chloride (Cl^-) in interstitial water were measured using an ICS-900 ion chromatography system (Thermo Scientific, USA). Sulfide was stabilized with zinc acetate and determined using an improved methylene blue colorimetric method [35]. The copy number of the *hgcA* gene (*hgcA*_261F: CGGCATCAAYGTCTGGTGYGC, *hgcA*_912R: GGTGTAGGGGGTGCAGCCSGTRWARKT) was determined to characterize the abundances of Hg-methylating microorganisms in soils by Majorbio, and detailed determination methods have been reported elsewhere [36].

2.2.2. Dissolved organic matter analyses

Dissolved organic matter (DOM) concentration, measured as dissolved organic carbon (DOC) concentration, was quantified using a TOC analyzer (GE Sievers InnovOx, USA). The structure of DOM was characterized by ultraviolet-visible absorption (UV-Vis) and excitation-emission-matrices (EEM) spectra, measured by an Aqualog® spectrophotometer (Horiba, France). Details of the determination method have been reported in previous studies [37–39] and are provided in the Supplementary Materials. In this study, we selected the absorption coefficient including α(355), $SUVA_{254}$, $SUVA_{260}$, and spectral slope ratio (S_R), and fluorescence parameters including fluorescent index (FI), humification index (HIX), and biological index (BIX) to analyze the DOM characterization. The relevant absorption and fluorescence parameters are presented in Table S3 and Table S4, respectively.

2.2.3. Mercury analyses

For water samples, THg was determined following EPA Method 1631 [40]. This method involves oxidation of all Hg species to Hg^{2+}, reduction to elemental Hg^0, preconcentration on gold traps, and final quantification by cold vapor atomic fluorescence spectroscopy (Tekran 2500, USA). MeHg analysis followed EPA Method 1630 [41], involving distilled, ethylated, gas chromatographically separated, and detection using MeHg Analytical System (Brooks Rand model III, USA).

For soil samples, THg was extracted by digestion with a mixture of aqua regia and BrCl, followed by quantification using a cold vapor atomic absorption spectrometry (F732-VJ, Shanghai). Soil MeHg was digested, extracted, re-extracted to obtain the extract [42]. MeHg in the extract was then derivatized with ethylation and quantified using an automated mercury analyzer (Tekran 2700, USA).

2.3. Quality assurance (QA) and quality control (QC)

To avoid contamination and maintain ultralow blanks, ultraclean benches, ultrapure water, disposable gloves and guaranteed reagents (GR) were employed during the laboratory analysis. Besides, laboratory quality control was implemented through blank experiments, parallel experiments, mass balance, and spike recovery. Throughout the analysis, continuous quality control was demonstrated by method blanks consistently below the detection limit and by standard reference materials (SRMs) recoveries, analyzed with each sample batch, all falling within the 80–120 % acceptance range (see Table S5 for specific detection limits and detailed recovery data).

2.4. Statistical analyses

Statistical analyses were conducted using Origin 2024b, SPSS 26.0, WPS Office 2021 and RStudio 4.4.2. All results were based on at least three replicate measurements to ensure reliability. One-way ANOVA tests were conducted to compare the means of three or more independent samples (treatments) simultaneously. Mantel test was used to explore the potential link between the soil/interstitial water MeHg (MeHg%) and soil properties, as well as DOM spectral indices. Pearson's correlation analysis identified the relevance between soil properties and DOM spectral indices. Then, partial least squares path modeling (PLS-PM) was employed to estimate the complex relationships between latent variables. Finally, the PHREEQC computational program was employed to thermodynamically simulate the speciation of Hg [43,44]. The detailed modeling process and the reaction equilibrium constants of Hg are provided in the Supplementary Materials (Table S6).

3. Results

3.1. Dynamics of soil Eh, pH and organic matter

The type of organic material directly governed the rate and intensity of anaerobic reduction, as reflected in distinct temporal redox patterns across treatments (Fig. 1**a**). System Eh values ranged from –29.33 ± 16.35 to –397.6 ± 12.08 mV, confirming sustained anaerobic environment. Rice straw and PM treatments rapidly established strongly reduced environments by day 10, achieving Eh values of –304.17 ± 1.65 mV and –350.93 ± 10.46 mV, respectively, obviously lower than those of other treatments ($p < 0.05$). These systems subsequently stabilized within –200 to –300 mV. In contrast, CD and CM required extended reducing equilibration, reaching minimum Eh of –237.3 ± 37.11 mV and –380.43 ± 11.35 mV by day 33, respectively. Interestingly, CK consistently maintained higher Eh values than other treatments throughout the incubation, suggesting that the addition of organic amendments promoted a decrease of Eh. Moreover, RS and PM showed significantly higher ΔEh (i.e., the difference between the treatment Eh and the control Eh) values compared to CD and CM in preculture period (day 10). However, by the mid-incubation period (days 33 and 46), CM exhibited a significantly higher ΔEh than others. By day 70, the Eh values of all treatments gradually stabilized.

The incorporation of organic materials induced systematic pH modulation in the incubation system (Fig. 1**b**). The pH values in RS and PM treatments were significantly lower than other treatments ($p < 0.05$). Specifically, RS and PM maintained a weakly acidic environment, while CD, CM and CK remained in a weakly alkaline condition. Over time, there was a general downward trend in pH across all treatments, and pH stabilization occurred near neutrality at approximately 7.0 by day 33.

The application of organic materials increased SOC concentrations across all treatments (Fig. S1), with concurrent increase in DOC exhibiting distinct temporal and material-specific patterns (Fig. 1**c**). Rice straw and PM treatments displayed rapid DOC accumulation, peaking at 315.87 ± 3.70 and 177.63 ± 1.05 mg/L on day 18 (2–10 and 2–6 times than CK, respectively), followed by progressive decline to stabilization by day 70. In contrast, CD exhibited significantly higher DOC concentrations than CK only on day 70 ($p < 0.05$), with no differences observed at other time points. Chicken manure consistently surpassed CK ($p < 0.05$), though its DOC remained relatively stable during incubation (48.37 ± 0.39–71.15 ± 5.68 mg/L).

The structural characteristics of DOM under different treatments are illustrated in Figs. 1d–f, S2 and S3. Excluding day 0, the fluorescence index (FI) values ranged from 1.4 to 1.9, indicating that the DOM has dual-source characteristics, with allochthonous and autochthonous contributions coexisting across treatments (Fig. 1**d**). Notably, from day 10 to day 70, the CM treatment had significantly higher FI values ($p < 0.05$), exhibiting a more prominent autochthonous-source characteristic. This was corroborated by the biological index (BIX), where CM showed significantly elevated values ($p < 0.05$) compared to other treatments (Fig. 1**e**), confirming its higher content of newly produced endogenous OM. Notably, the humification index (HIX) values of DOM

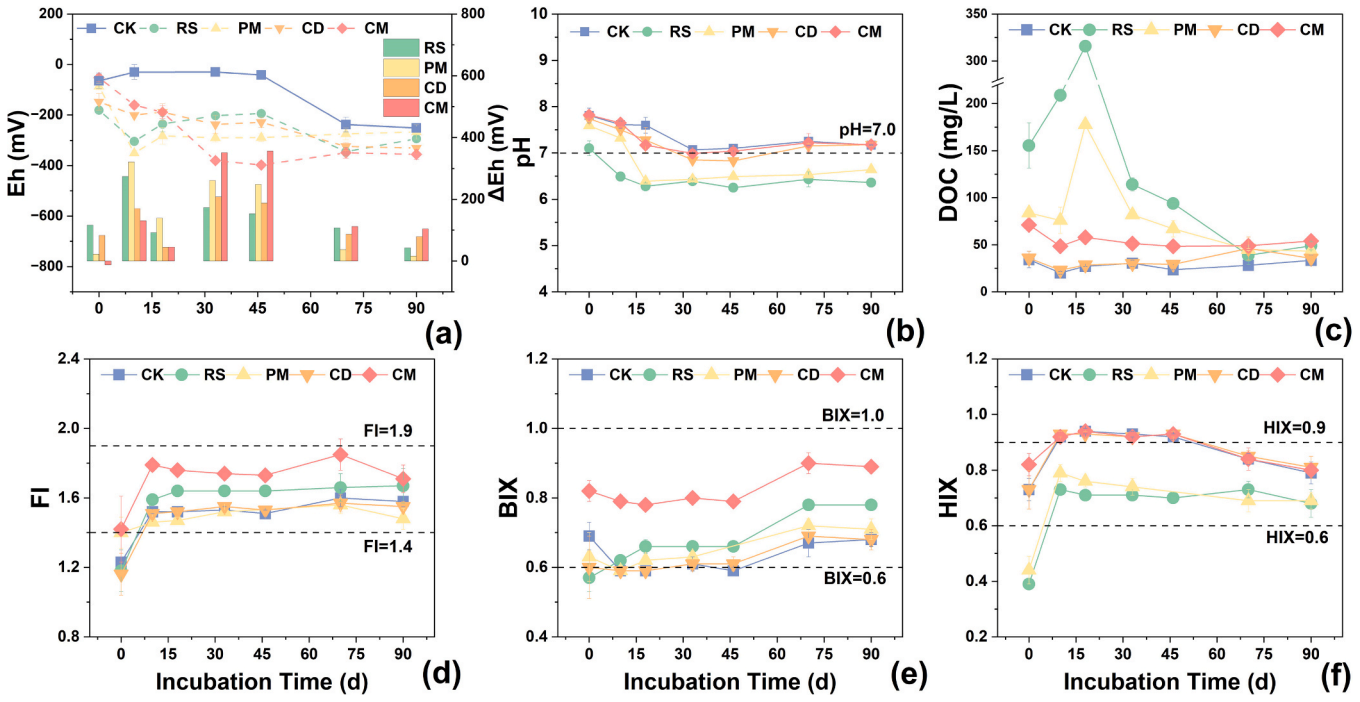


Fig. 1. (a) Eh and ΔEh (relative to CK); (b) pH; (c) DOC concentrations; (d) FI; (e) BIX; (f) HIX of interstitial water for five treatments (CK, RS, PM, CD, CM) at seven incubation time points. Data are presented as the mean ± standard deviation ($n = 3$).

in CK, CD, and CM treatments were significantly higher ($p < 0.05$) than PM and RS treatments (Fig. 1**f**), demonstrating the presence of more recalcitrant humic substances. Despite high autochthonous FI/BIX signals in CM treatments, its elevated HIX implies that its DOM retained stable humified components. Furthermore, the fluorescence composition differed among treatments: RS and PM were characterized by peaks A (humic-like components) and B (protein-like components), CD and CK by peak A, and CM primarily by peak M (humic-like components) (Fig. S3).

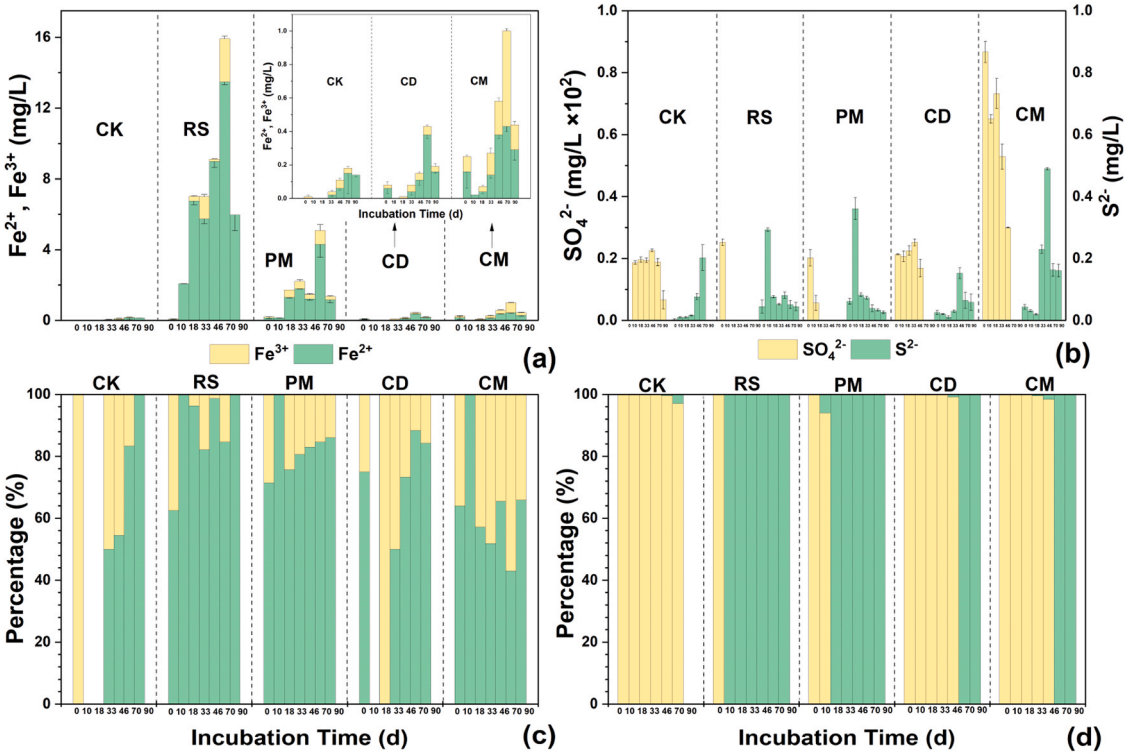


Fig. 2. Dynamics of (a) Fe^{2+} and Fe^{3+} concentrations, (b) SO_4^{2-} and S^{2-} concentrations, (c) the Fe^{2+}/Fe^{3+}ratio, and (d) the SO_4^{2-}/S^{2-} ratio in the interstitial water among the five treatments (CK, RS, PM, CD, CM) over the incubation period.

3.2. Variations in Fe and S in interstitial water

The amendment of organic materials promoted the reductive dissolution of Fe(III) oxides and enhanced sulfate reduction, driving different Fe-S transformation dynamics in the incubation system (Fig. 2). Ferrous iron emerged as the predominant species in the interstitial water across all treatments (Fig. 2**c**). In particular, RS and PM exhibited significantly enhanced Fe^{2+} production, peaking at 13.48 ± 0.15 and 4.30 ± 0.73 mg/L on day 70, which were significantly higher ($p < 0.05$) than those observed in CD and CM (Fig. 2**a**).

Sulfate reduction also revealed treatment-specific patterns. Sulfate became undetectable in RS by day 10 and PM by day 18, whereas CM and CD maintained detectable SO_4^{2-} until day 70 (Fig. 2**b**). Subsequent complete sulfate depletion coincided with nearly 100 % sulfide predominance in interstitial water in all treatments (Fig. 2**d**). In addition, sulfide concentration showed an initial increase followed by a gradual decline over time. The sulfide concentrations of RS and PM peaked on day 10 at 0.29 ± 0.006 and 0.36 ± 0.04 mg/L, followed by decrease and stabilization, while the CM and CD exhibited a noticeable increase in sulfide concentration on day 33, reaching their peaks of 0.49 ± 0.004 and 0.15 ± 0.02 mg/L on day 46, before stabilizing on day 70.

3.3. Dynamics of mercury

3.3.1. Dynamics of THg concentrations

The interstitial water THg (WTHg) concentrations ranged from 0.45 ± 0.03–7.73 ± 0.05 ng/mL (Fig. S4). A decline in WTHg levels was observed across all treatments by day 10, after which concentrations stabilized. Notably, RS maintained significantly lower WTHg concentration compared with other treatments ($p < 0.05$) across all sampling intervals except day 0.

3.3.2. Temporal patterns of MeHg production

As depicted in Fig. 3, the amendment of organic materials increased the MeHg concentrations both in soil and interstitial water to varying degrees compared to CK. During initial incubation stages, RS and PM treatments exhibited pronounced MeHg accumulation in soil. Specifically, the RS treatment achieved its peak soil MeHg (SMeHg) level (7.51 ± 1.16 ng/g) on day 10, representing 24.2-fold higher than CK, while PM treatment peaked later on day 18 with SMeHg concentration of 3.94 ± 0.32 ng/g, reflecting 13.6-fold increase over CK. In contrast, CM displayed delayed Hg methylation dynamics, reaching its SMeHg maximum (12.94 ± 0.88 ng/g) on day 70, which was 14.5-fold higher than CK. Notably, the CD showed no statistically significant SMeHg differences from CK throughout the incubation period (Fig. 3**a**).

The variations in interstitial water MeHg (WMeHg) concentrations followed a pattern similar to those observed in SMeHg (Fig. 3**b**). Both RS and PM treatments reached peak WMeHg concentrations on day 10 (265.51 ± 4.53 pg/mL and 58.80 ± 1.51 pg/mL, respectively), representing 390.5-fold and 86.5-fold greater than CK. The CM treatment reached its highest WMeHg concentration on day 70 (158.63 ± 7.15 pg/mL), showing 19-fold higher than CK. In comparison, the CD treatment peaked on day 46 with a WMeHg concentration of 19.82 ± 7.27 pg/mL, marking a 5.3-fold higher than CK. Following peak concentrations, WMeHg levels in all treatments exhibited a gradual decline, yet remained consistently higher than CK.

The proportion of MeHg to THg (MeHg%) is a widely used indicator to characterize the degree of Hg methylation potential [45,46]. In this study, during the early incubation stages, the soil/water Hg methylation potentials in RS and PM were relatively high. However, by the later

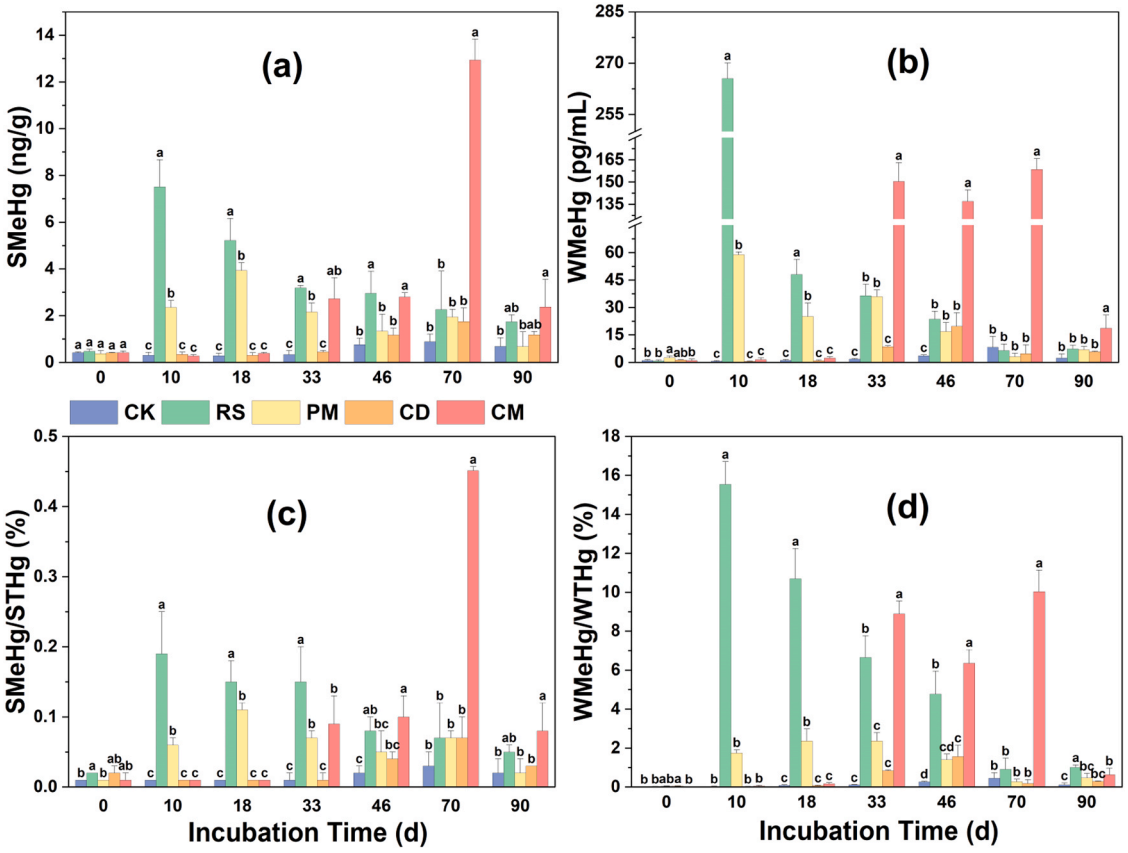


Fig. 3. Temporal variations in (a) soil MeHg (SMeHg), (b) interstitial water MeHg (WMeHg), (c) SMeHg/STHg ratio, and (d) WMeHg/WTHg ratio across the five treatments (CK, RS, PM, CD, CM) during the incubation period. The different letters on the bar chart represent the significant differences among treatment groups at the same time (one-way ANOVA test, $p < 0.05$). Data are shown as the mean ± standard deviation ($n = 3$).

stages, Hg methylation potential in CM was the most pronounced among the four treatments. Over time, the soil/water MeHg% concentrations showed a pattern of initial increase followed by a decline and ultimately stabilized in each treatment (Fig. 3**c and d**).

The *hgcA* gene is a specific and essential marker for known Hg-methylating microorganisms, and its copy number serves as a robust proxy for the functional potential of these microbes [36,47]. Quantification of *hgcA* gene in paddy soil on days 0, 18, and 48 revealed that its abundance increased over time and was markedly elevated by organic amendments relative to the control ($p < 0.05$) (Fig. S8).

3.3.3. MeHg solid-liquid partitioning

The apparent solid-liquid partitioning ($\mathrm{Log}K_d$) for MeHg was calculated as the logarithm of the ratio of its concentration in the solid phase to that in the aqueous phase, reflecting phase distribution capacity of MeHg under different organic material treatments [48–50]. Throughout incubation, $\mathrm{Log}K_d$ values ranged from 1.24 ± 0.17–2.93 ± 0.36 L/kg across different treatments (Fig. S5). In the pre-incubation period (day 10), $\mathrm{Log}K_d$ values in RS and PM were significantly lower than those in other treatments ($p < 0.05$), indicating enhanced aqueous partitioning. During the mid-incubation period (days 33 and 46), MeHg $\mathrm{Log}K_d$ values in all amended systems were significantly lower than CK ($p < 0.05$), confirming that organic amendment facilitated MeHg migration to the liquid phase. However, by day 90, no significant differences were observed among the different treatments.

4. Discussion

Our findings indicate that the addition of agricultural organic materials significantly enhanced the potential for Hg methylation in paddy soils, with the magnitude of this enhancement modulated by the decomposition rate and degree of each organic material. To elucidate the underlying mechanisms governing MeHg dynamics, we systematically evaluated the interdependent transformations of key biogeochemical parameters (including pH, Eh, DOM and Fe/S species), and identified their specific contributions to Hg methylation potentials in simulated paddy systems under various organic materials applications.

4.1. Divergent changes in critical soil properties induced by different organic amendments

The application of organic materials significantly increased the SOC pool (Fig. S1), triggering distinct biogeochemical transformations [23, 51] whose timing and extent varied considerably with the chemical composition and nature (e.g., C/N ratio, humification degree) of the materials applied. The high C/N ratios in RS (C/N = 32) and in the straw-incorporated PM (C/N = 45), a material whose biogeochemical behavior aligned more closely with that of RS (Table S2), created a stoichiometric imbalance that triggered microbial secretion of extracellular enzymes to accelerate the decomposition of native SOM, a nitrogen acquisition strategy that induced short-term DOC accumulation (Fig. 1**c**) [52,53]. Concurrently, this decomposition process generated reducing metabolites such as organic acids and CO_2, decreasing the pH (Fig. 1**b**), and driving Eh below –300 mV within 10 days (Fig. 1**a**). Correlation analysis **(**Fig. S6) further revealed a significant negative relationship between Eh and SOC content during the incubation period ($p < 0.01$), confirming the close association between SOC accumulation and Eh reduction. Under these strongly reducing conditions, enhanced microbial electron transfer facilitated rapid Fe(III) and sulfate reduction, reflected by pronounced increases in Fe^{2+} and S^{2-} concentrations (Fig. 2). These observations are consistent with previously reported reductive conversion of elemental sulfur and sulfate to sulfide under anoxic conditions (Eh = –200 mV), accompanied by iron speciation dominated by Fe^{2+} [54]. The decline in the S^{2-}/Fe^{2+} molar ratio below 1.0 by day 18 (Fig. S7) likely suggests rapid FeS precipitation, mechanistically explaining the subsequent decreases in dissolved S^{2-} and Fe^{2+} concentrations (Fig. 2).

In contrast, CD (C/N = 5) and CM (C/N = 9) exhibited lower C/N ratios (Table S2), promoting preferential microbial utilization of inherent C and N nutrients. This resulted in minimal SOM decomposition and negligible DOC accumulation (Fig. 1**c**). Notably, CM characterized by high OM and nitrogen content (Table S2), simulated more active microbial metabolism, as evidenced by significantly higher BIX values compared to other treatments (Fig. 1**e**). However, the identical CM material used herein was independently characterized as highly humified [18], reflecting a substantial recalcitrant carbon pool that contributed to a delayed release of bioavailable carbon and electrons. Consequently, a pronounced enhancement in iron-sulfur reduction was postponed until the late-phase incubation (> 33 days), accompanied by a decrease in S^{2-}/Fe^{2+} molar ratios below 1.0 (Fig. S7). The CK treatment sustained fundamental microbial activity and stable DOC concentrations in the absence of organic amendments, whereas the limited organic input of CD resulted in DOM signatures statistically indistinguishable from CK ($p > 0.05$). As incubation proceeded, depletion of labile carbon shifted microbial metabolism toward recalcitrant organic fractions. Concurrently, DOC concentrations decreased (Fig. 1**c**), largely driven by microbial consumption as an energy source and the concomitant mineralization of OM [55]. This sequential consumption of labile DOC played a pivotal role in regulating the observed Fe/S dynamics: it first fueled rapid Fe(III) reduction, and as labile carbon was depleted, it supported the subsequent surge in sulfate reduction [54,56]. Ultimately, the depletion of these readily available carbon pools reduced the microbial regeneration of Fe^{2+} and S^{2-}, promoting their precipitation as Fe-S minerals and leading the system toward redox equilibrium. All systems reached redox equilibrium by day 70, indicated by the stabilization of Eh values and Fe/S species concentrations (Figs. 1 and 2).

Interestingly, temporal patterns of Fe-S reduction and Fe(II)/sulfide generation were strongly coupled with Eh dynamics across all systems (Figs. 1 and 2). Correlation analyses revealed significant positive (SO_4^{2-} and Eh) and negative (Fe^{2+} and Eh) relationships (Fig. S6), demonstrating that exogenous carbon-induced Eh shifts serve as the primary regulator of iron-sulfur cycling. Moreover, high C/N organic materials facilitated faster establishment of iron-sulfur buffering systems, which may subsequently influence Hg methylation processes [25,57,58].

4.2. Organic amendment-induced enhancement of Hg methylation potentials in paddy soils

4.2.1. Relative contributions of diverse soil properties

The application of various organic materials enhanced Hg methylation potentials to varying degrees and at different time points. Notably, the temporal patterns of MeHg concentrations closely tracked the dynamics of key soil indicators such as DOM, Eh and Fe/S (Figs. 1, 2, and 3), suggesting organic materials addition facilitates Hg methylation potentials by altering complex soil properties.

Previous studies have generally emphasized the role of increased DOC concentration in promoting Hg methylation following organic materials input [22,23,59]. Nevertheless, our findings demonstrated no significant statistical correlation between DOC concentration and MeHg (Mantel's $p > 0.05$ and Mantel's $r < 0.4$), except for SMeHg in PM and CD treatments (Fig. 4). Instead, DOM spectral parameters exhibited a stronger association with MeHg production, especially in RS treatment (Fig. 4), indicating that DOM quality, rather than its concentration, plays a more critical role in mediating Hg methylation [38]. This view is supported by correlation analyses, which revealed a significant positive correlation between FI/BIX and MeHg, as well as a negative correlation between $SUVA_{254}/SUVA_{260}$ and MeHg **(**Fig. S6**).** These findings suggest that DOM with lower aromaticity and higher bioavailability, probably acting as electron donors or labile carbon sources, can enhance the abundance and activity of Hg-methylating microorganisms, thereby promoting Hg methylation [60,61]. This interpretation is further corroborated by the notable increase in *hgcA* gene copy numbers after

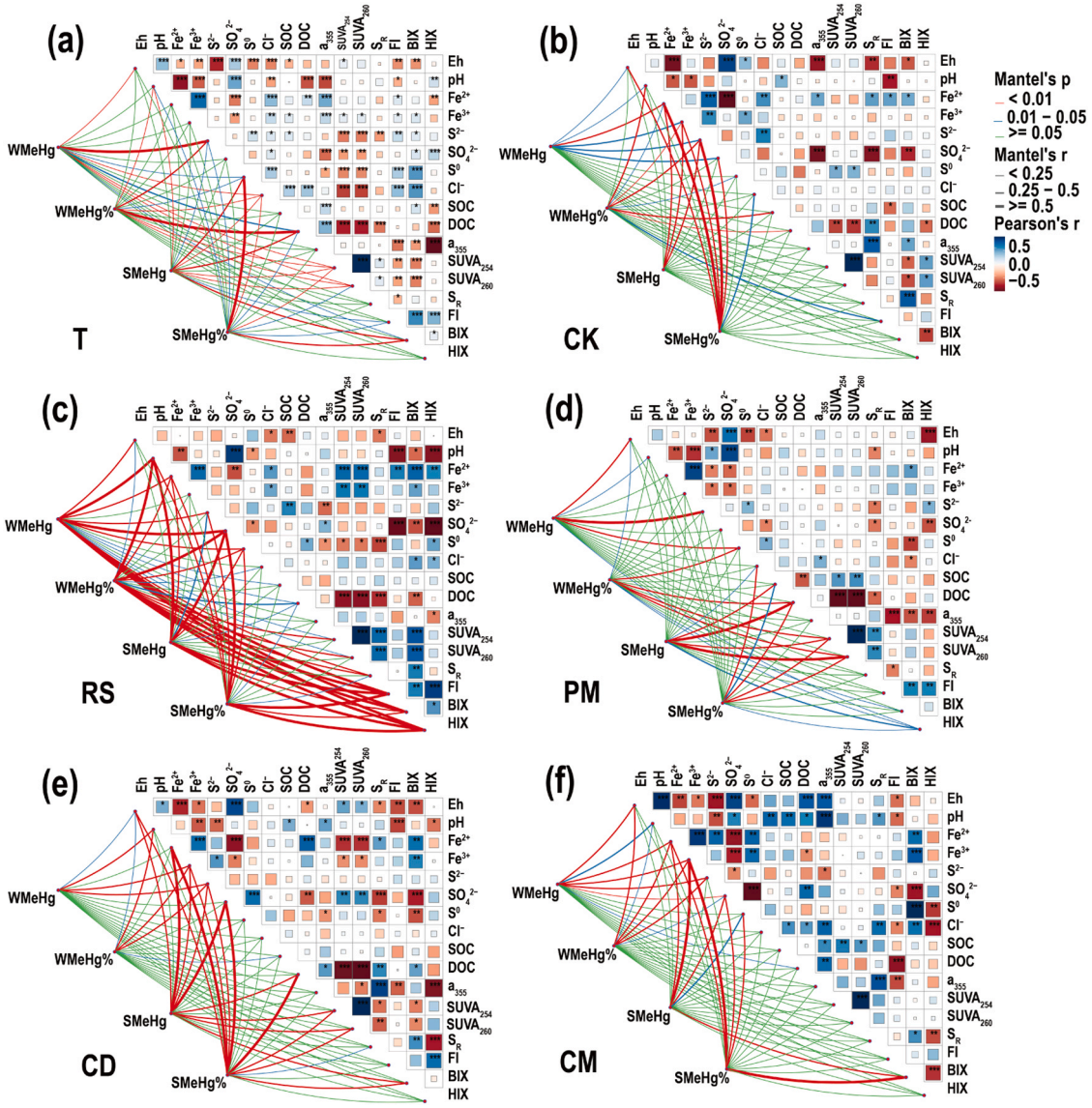


Fig. 4. Relationships between soil/interstitial water MeHg (MeHg%) and soil properties/DOM spectral indices for (a) All treatments (T), and individual treatments (b) CK, (c) RS, (d) PM, (e) CD, (f) CM. Pairwise Pearson correlations between environmental variables are shown by a color gradient (blue for positive, red for negative). Asterisks represent significant effects: *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$. Mantel test results between environmental variables and Hg variables (SMeHg, SMeHg%, WMeHg, WMeHg%) are represented by connecting lines, with width proportional to Mantel's r statistic and color indicating significance (blue: $p < 0.05$, red: $p < 0.01$).

organic material addition (Fig. S8), indicating an elevated potential for microbial methylation of Hg.

Although pH is known to influence Hg bioavailability by modulating soil particle exchange and organic matter composition [46,62], no significant correlation between pH and MeHg was observed in this study (Figs. 4 and S6), suggesting that pH changes are not a primary driver of MeHg dynamics in this study. Similarly, although Cl^- concentration increased due to chloride input from organic amendments (particularly RS and PM), it was also not identified as a critical factor affecting Hg methylation (Figs. S6 and S9). Notably, Mantel tests (Fig. 4) indicated a strong association between Eh and MeHg production. The synchronous temporal dynamics, where lower Eh corresponded to higher MeHg concentrations (Figs. 1 and 3), support the hypothesis that Hg methylation potential is closely linked to the rapid establishment of anaerobic conditions after organic material application. Simultaneously, low redox potentials favored the formation of Fe-S reduction products that alter Hg speciation and adsorption/desorption behavior, thereby facilitating Hg methylation [7,63]. Furthermore, Mantel tests demonstrated that Fe/S ions significantly affected MeHg production (Fig. 4). Taken together, these results suggest that the Fe/S cycle, modulated by decreasing Eh, serves as a dominant driver of Hg methylation potentials in paddy fields.

4.2.2. *Dominant role of iron and sulfur reduction driven by Eh decline*

To further explore the key mechanisms regulating Hg methylation potentials following organic material application, we employed PLS–PM (Fig. 5). The results revealed that Eh indirectly impacted MeHg production through its effect on sulfide concentrations in the RS, PM and CD treatments (path coefficients: –0.36, –0.58 and –0.31, respectively), with sulfide further exerting a direct positive effect on MeHg levels (path coefficients: 0.70, 0.45 and 0.76, respectively). This highlights the significance of sulfur biogeochemical cycling in Hg methylation potentials in paddy soils [26,54]. Additional support came from the observed increase in S^0, an intermediate in inorganic sulfide transformation, during sulfate reduction (Fig. S10), which correlated positively with MeHg in RS and PM (Fig. S6). S^0 can react with HS^- to form polysulfides, which strongly complex with Hg^{2+}, promoting the dissolution of insoluble HgS,

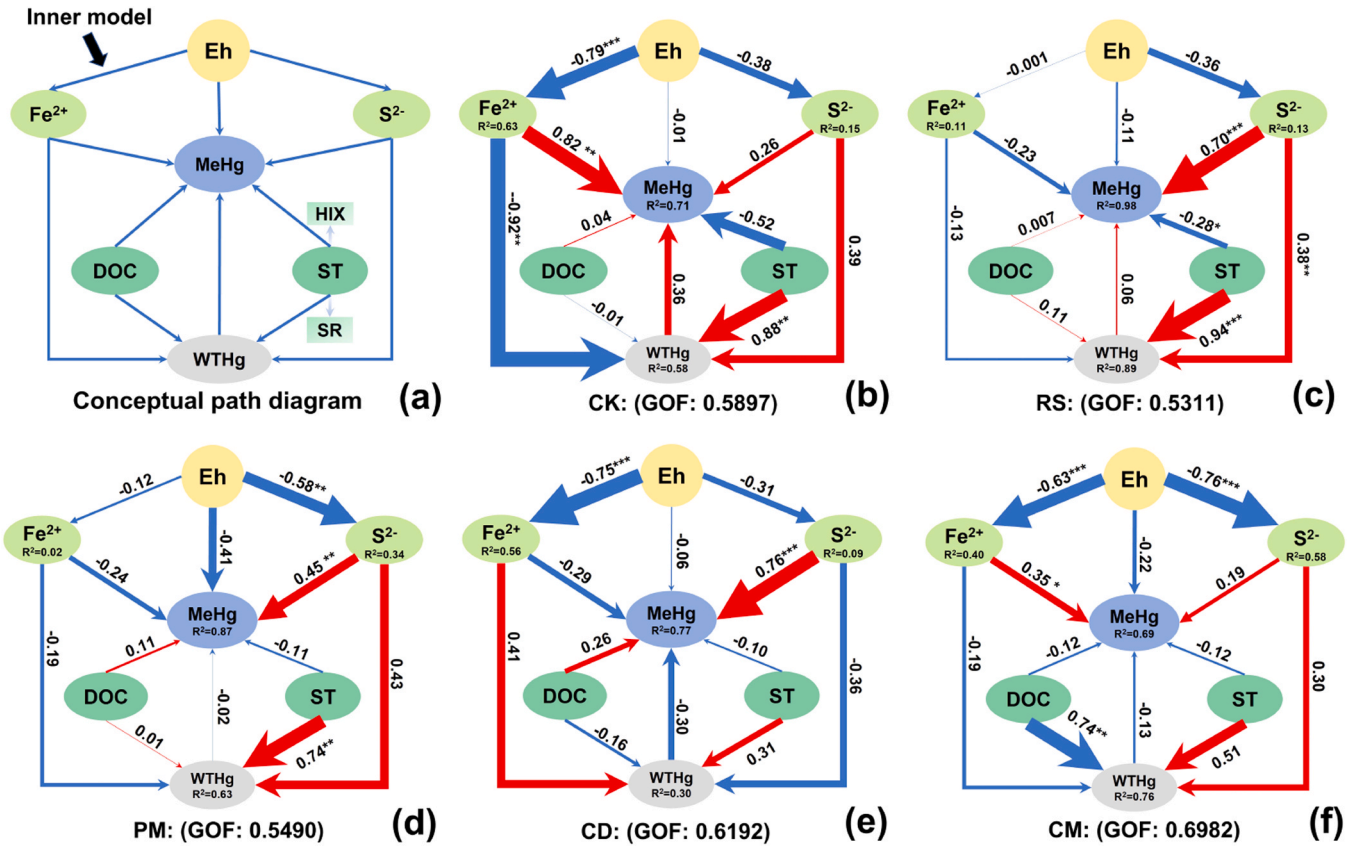


Fig. 5. Results of the PLS-PM models for MeHg. (a). Conceptual path diagram depicting the cause-effect relationships of different soil properties and MeHg concentrations. The reflecting indicators of MeHg is interstitial water MeHg (WMeHg). DOM structure (ST) is represented by HIX and S_R. (b)~(f) Model results for each treatment: CK, RS, PM, CD, and CM. Arrow colors denote path directions (red: positive, blue: negative), widths are proportional to standardized coefficients (values shown on paths), and R^2 values indicate explained variance. Model goodness-of-fit is indicated by the GOF index. Significance levels: *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$.

and increasing the pool of bioavailable Hg for methylation [28,64,65]. Consistent with this, PHREEQC modeling predicted the predominance of soluble Hg-sulfide complexes (HgS_2^{2-}, $HgHS_2^-$, and $Hg(SH)_2^0$) under these conditions (Fig. S11), indicating the formation of soluble Hg polysulfide species facilitated by HgS dissolution in the presence of sulfide and organic materials amendment. Besides, MeHg production in CK and CM was primarily mediated by Fe^{2+} (path coefficients: 0.82 and 0.35, respectively), with Eh influencing MeHg levels via Fe^{2+} (path coefficients: –0.79 and –0.63, respectively). This suggests that under lower Eh conditions, Fe(III) reduction become a dominant pathway facilitating Hg methylation, likely through the reductive dissolution of Fe(III) oxides and the release of adsorbed IHg, thereby enhancing its bioavailability [54,66].

Notably, the observed initial increase and subsequent decline in sulfide and Fe(II) concentrations (Fig. 2) can be attributed to Eh reduction driven by the degradation of organic materials, which rapidly depletes electron donors and promotes the reduction of sulfate to sulfide and Fe(III) to Fe(II). This is followed by the precipitation of amorphous or weakly crystalline FeS [58,67–69]. Previous studies suggested that FeS can reduce Hg bioavailability via co-precipitate with Hg to form Hg-FeS complexes [63,70], or through displacement of Fe(II) by Hg(II) to form thermodynamically stable HgS [68,71,72]. These processes likely reduced Hg bioavailability, which in turn suppressed Hg methylation. Consequently, the observed net decrease in MeHg during later incubation (Fig. 3) may be attributed to this inhibition of production, along with potential MeHg degradation over time [73,74].

Additionally, decreased LogK_d following organic materials application (Fig. S5) suggests enhanced transportation of MeHg from soil to the interstitial water [75]. This mobile fraction is more readily absorbed by rice roots, leading to MeHg accumulation in grains and elevating human exposure risks [23,54]. Overall, the addition of organic materials promotes Hg methylation potentials in paddy soils through multiple pathways. Beyond the established role of DOM, this study highlights the crucial mediating role of iron-sulfur cycle mediated by Eh decrease.

5. Conclusion

This study showed that organic materials characteristics govern Eh dynamics, which in turn drive Fe-S reduction that enhances Hg bioavailability and stimulates Hg-methylating microorganisms, leading to enhanced MeHg formation. Furthermore, we identified a ‘two-phase’ pattern in MeHg generation: an early promotion phase driven by iron-sulfur reduction, followed by a late suppression phase due to FeS precipitation and nutrient exhaustion. Notably, we demonstrate that decomposition dynamics of organic amendments are the key determinants shaping the distinct MeHg production patterns. This mechanistic understanding carries significant environmental implications. It reveals a critical time window for mitigating MeHg risk through strategic organic management. By selecting amendments with appropriate decomposition dynamics and optimizing application timing to avoid peak Hg methylation phases, we can effectively reduce grain MeHg accumulation. This approach fosters synergy between safeguarding food safety and sustaining the benefits of organic practices for soil health and crop productivity. Future studies employing techniques such as *hgcA*-targeted sequencing and metagenomics are warranted to directly link the established distinct redox environments to the composition and activity of specific Hg-methylating microbial communities under organic amendment, thereby refining risk predictions.

Environmental implication

The application of organic amendments, while beneficial to soil fertility and crop yield, can elevate the production of neurotoxic methylmercury (MeHg) in paddy soils. Our study revealed that the extent and timing of MeHg production varied depending on amendment-specific and primarily governed by Eh-driven iron-sulfur reduction. These findings offer a critical time window for risk mitigation through optimized organic materials management. By selecting organic materials with appropriate decomposition characteristics and optimizing application timing, and implementing targeted water management to regulate Eh, we can effectively reduce grain MeHg accumulation while sustaining soil health and crop productivity.

CRediT authorship contribution statement

Juan Wang: Writing – original draft, Investigation, Funding acquisition, Formal analysis, Data curation. **Jingwen Yang**: Methodology, Investigation, Conceptualization. **Yuping Xiang**: Visualization, Methodology, Conceptualization. **Jinping Xue**: Writing – review & editing, Visualization, Formal analysis. **Xiaosong Tian**: Writing – review & editing, Funding acquisition. **Tao Jiang**: Methodology, Funding acquisition. **Yongmin Wang**: Writing – review & editing, Visualization, Supervision, Funding acquisition. **Dingyong Wang**: Visualization, Validation, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jhazmat.2026.141582.

Data Availability

Data will be made available on request.

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